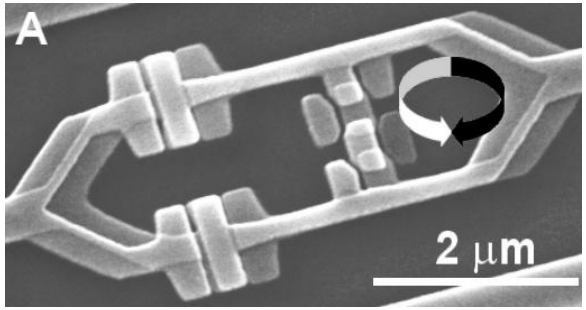
Quantum Algorithms

$$|188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059\rangle$$



Past, Present, and Future

$$|3980750864240649373971255005503864911990643623425267084063851895759463889572617685833317\rangle \otimes |4727721461074353025362230719730482246329146953020971164598521711305207112563635903975 27\rangle$$

Dave Bacon

Santa Fe Institute

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All lectures are given in the Robert N. Noyce Conference Room at the Santa Fe Institute unless noted.

Future CEPI Lectures

Date	Time	Speaker	Topic
11/10/04	reception: 4:15-5 p.m. talk: 5-6 p.m.	Wim van Dam University of California, Santa Barbara	T.B.A.

Internet

Quantum Algorithms

Do Physicists Make Good Quantum Programmers?



Dave Bacon
Santa Fe Institute



Computation and Science



Alan Turing



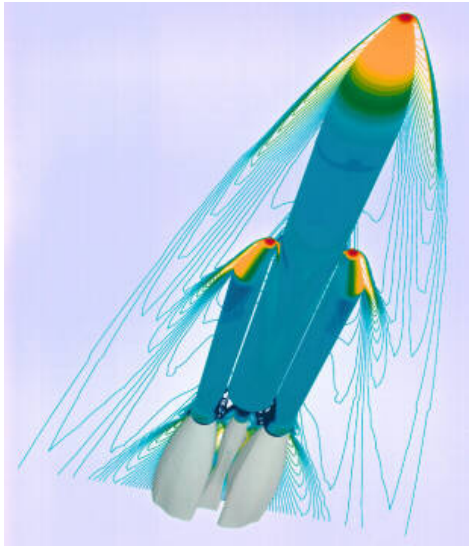
John von Neumann

Blue state

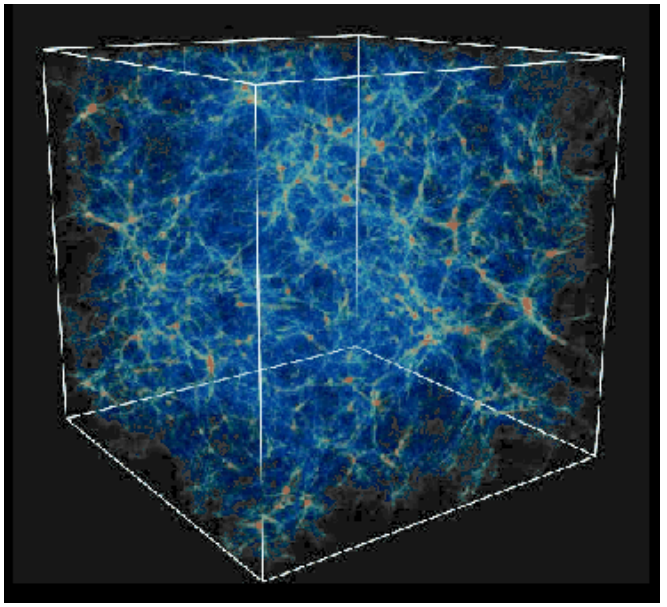
Red state

The invention of the modern computer changed the way we do science.

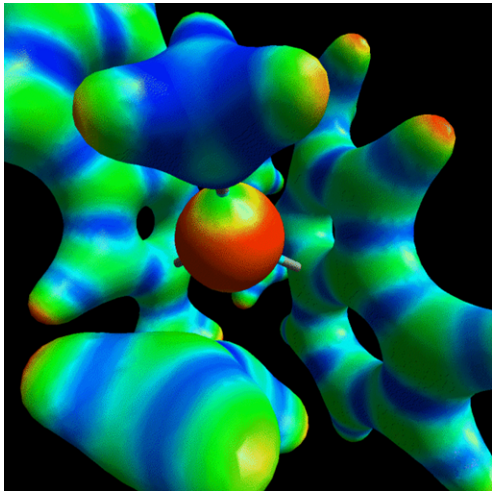
Simulation Revolution



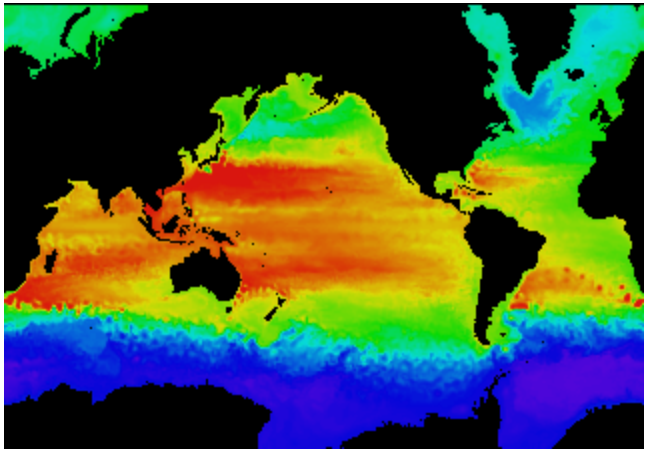
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Bryan and Norman (UIUC, NCSA, GC3)

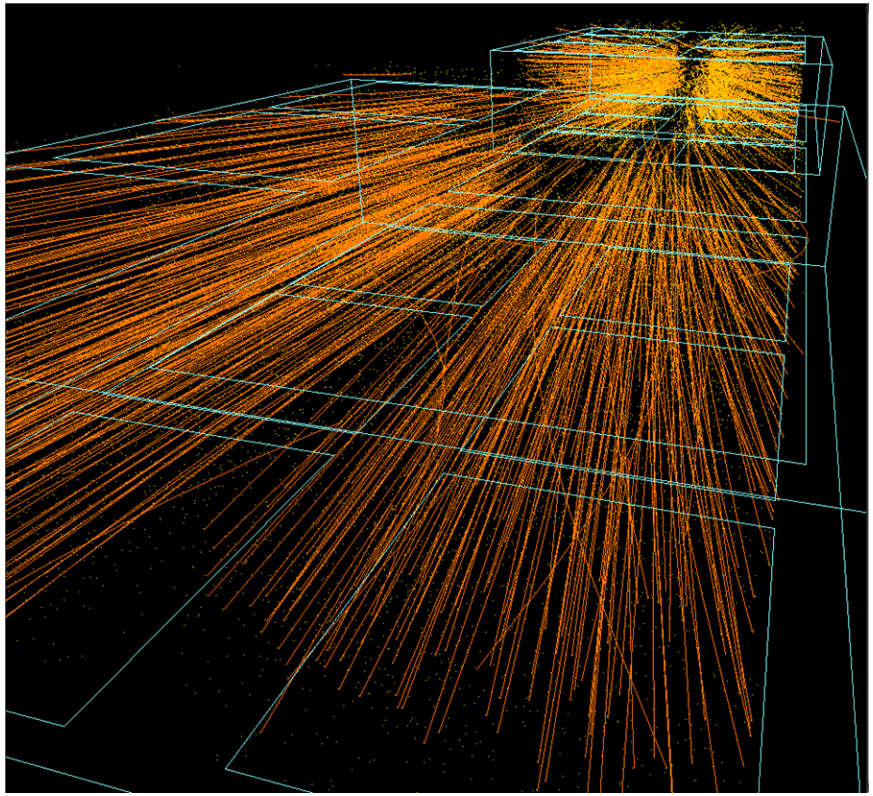


Rene Windiks, Bernard Delley (PSI)

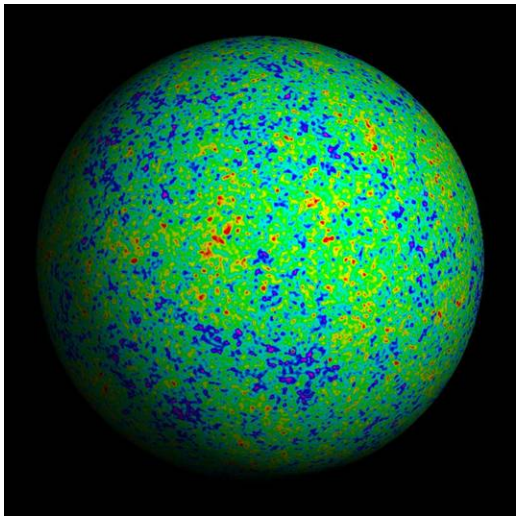


Computer Hardware, Advanced Mathematics and Model Physics (DOE)

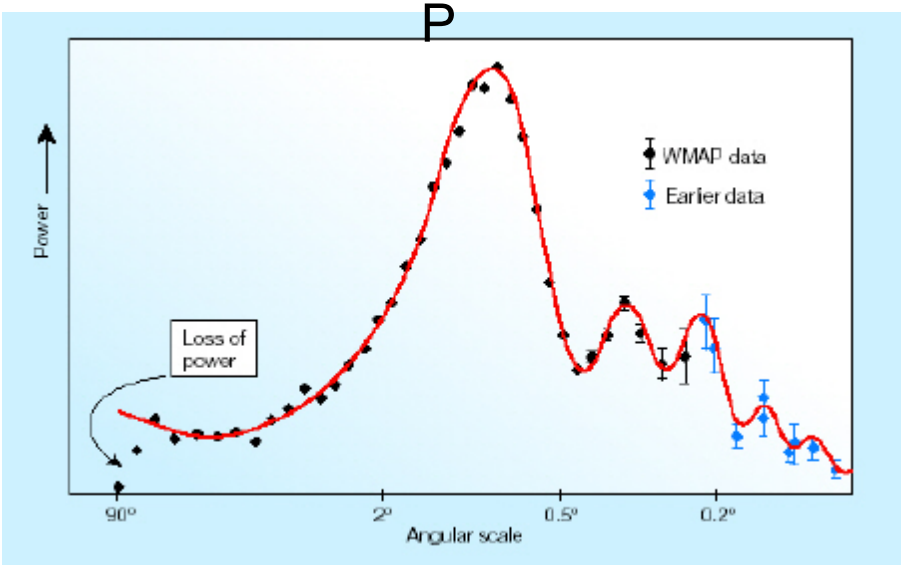
Data Revolution



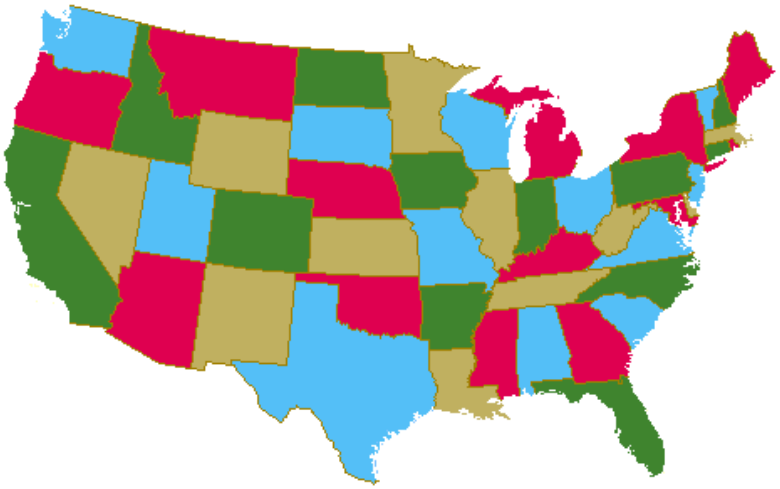
Cern NA49 experiment



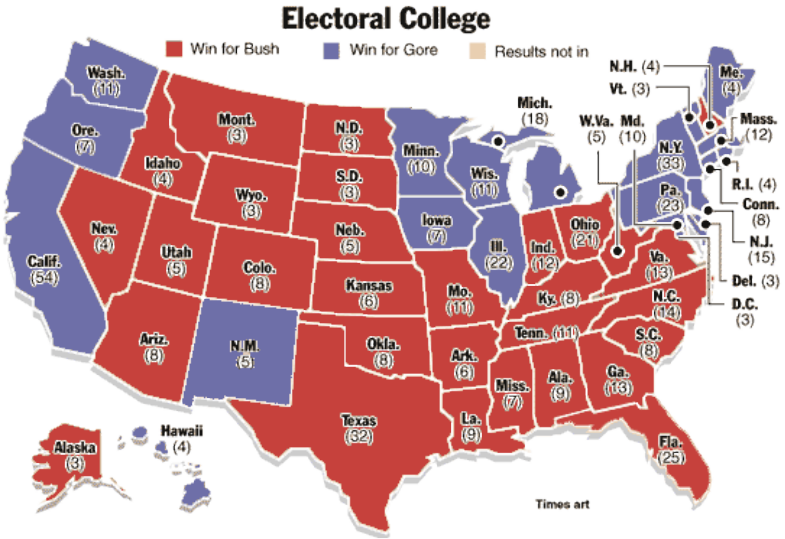
WMAP



Symbolic Revolution



Four Color Theorem



Two Color Reality

Not Our Universe

Suppose: a computer could search an unordered list of size N in time log(N) (strong P=NP world)



N

Then:

•Find a short proof of the Reimann hypothesis (unemployment for mathematicians?)

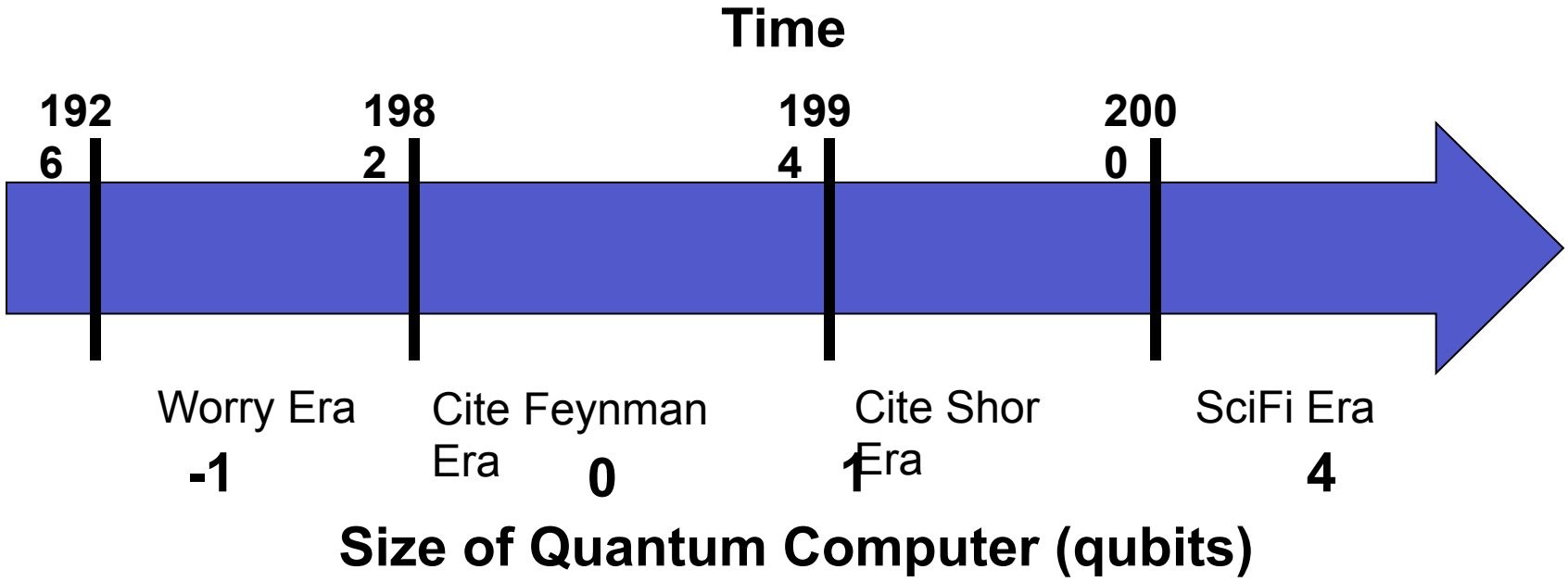
• Find the smallest polynomial time program for reproducing a data set (unemployment for theorists?)

• Perform near perfect pattern recognition, language comprehension, and many learning tasks (unemployment for humanity?)

Our Universe

“NATURE ISN'T CLASSICAL, damn it, and if you want to make a simulation of nature, you'd better MAKE IT QUANTUM MECHANICAL, and by golly it's a wonderful problem because it doesn't look easy.” – Richard Feynman

Quantum Computer: a computer whose evolution obeys the laws of quantum theory.



Quantum Science

Will quantum computation change the way we perform science?

<u>What</u> can a quantum computer do?

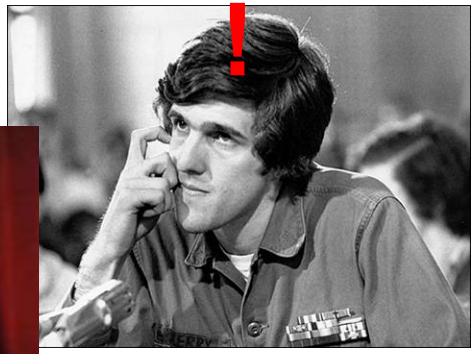
<u>What</u> good is a physicist?

A Game

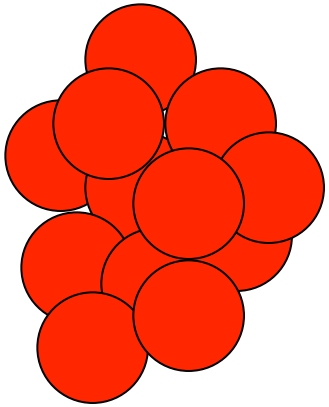
winner !

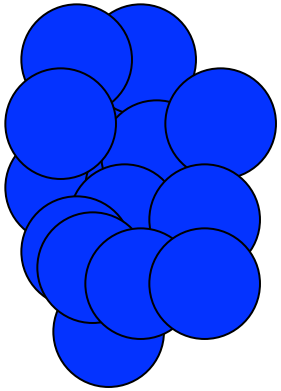


winner !









winner is last to place disk

The Physicist's Game

Mmmm…Symmetry



What are physicists good at?

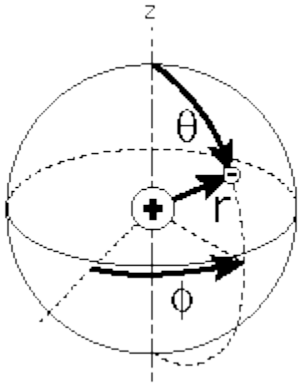
symmetry

symmetry

Why are physicists good at this?

1. **Nature is often symmetric!**
2. **Symmetric problems are solvable problems!**

Example: Hydrogen atom

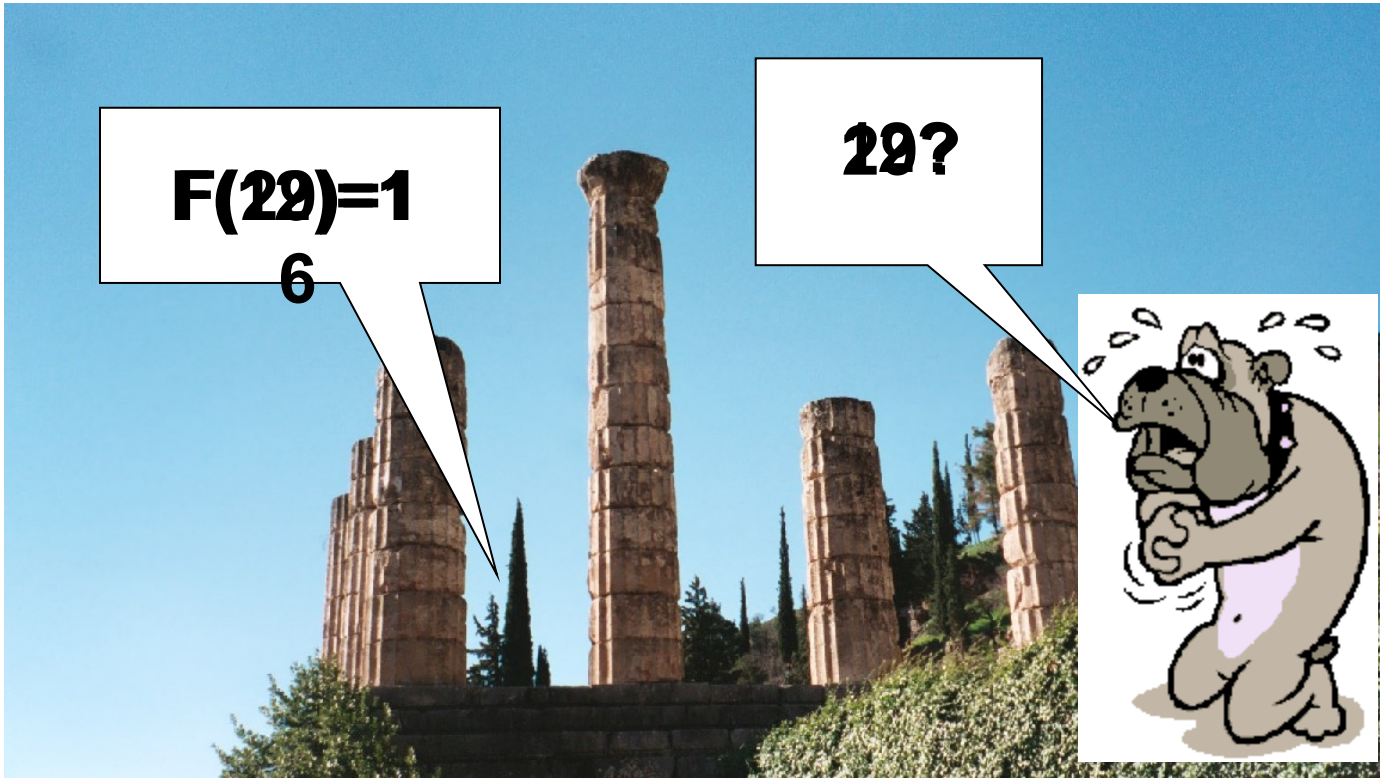


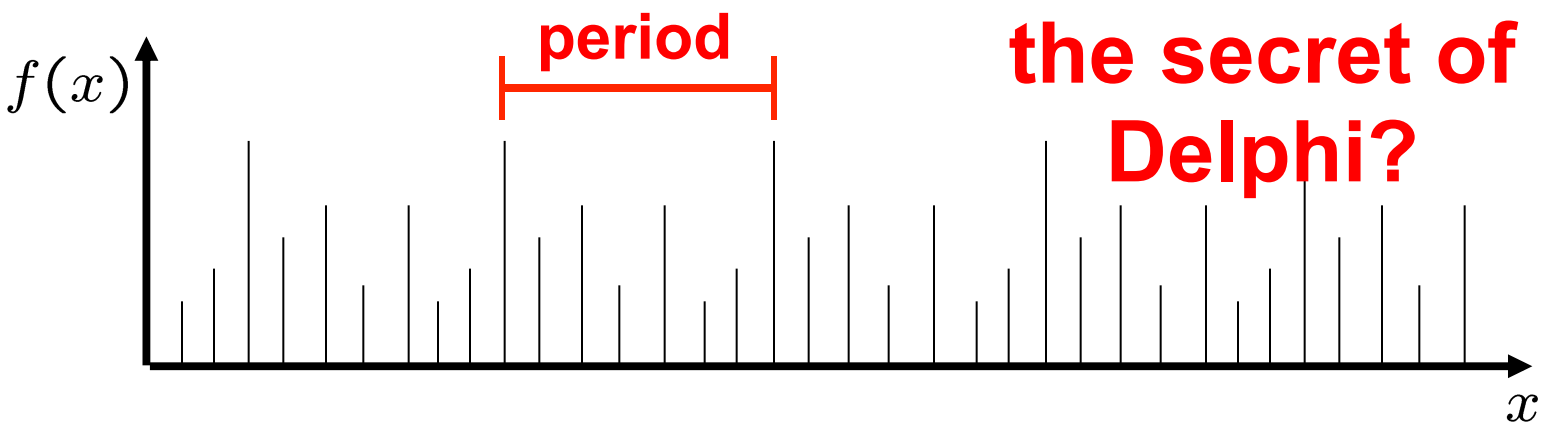
$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{r}\right]\psi(x,y,z) = E\psi(x,y,z)$$

spherical symmetry

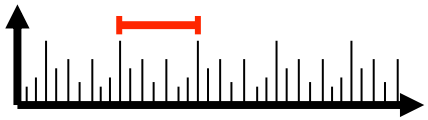
$$-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{e}{r} - E\right]R = 0$$

A Guessing Game

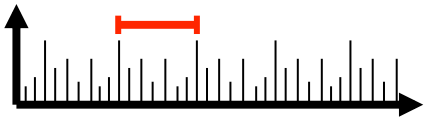




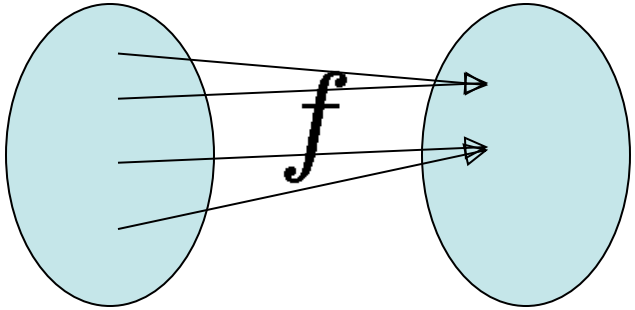
the secret of Delphi?



Period Finding



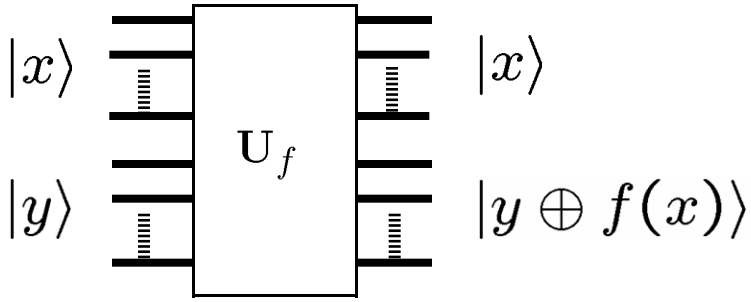
$x \in \{0,1\}^n$ $y \in \{0,1\}^n$



$$f(x) = f(x + l)$$

Problem: find ▌ in as few queries as possible

quantum oracle



$x, y, f(x) \in \{0,1\}^n$

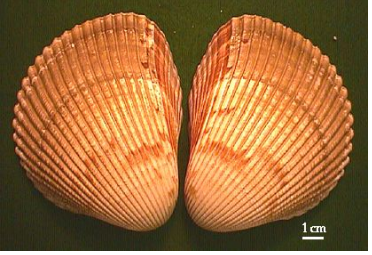
….in as few uses of the quantum oracle as possible

Period Finding Problem

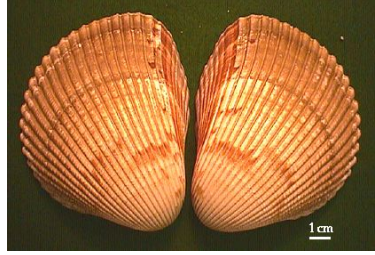
computer sciencese:

$\{0,1\}^n = \{0, \ldots, 2^n - 1\}$

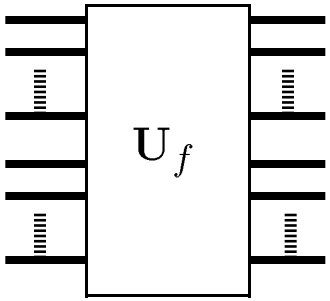
a symmetric problem!



Explicit Symmetry



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle \qquad |0\rangle$$



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle\otimes|f(x)\rangle$$

measure second register, randomly obtain $f(x_0)$

first register:

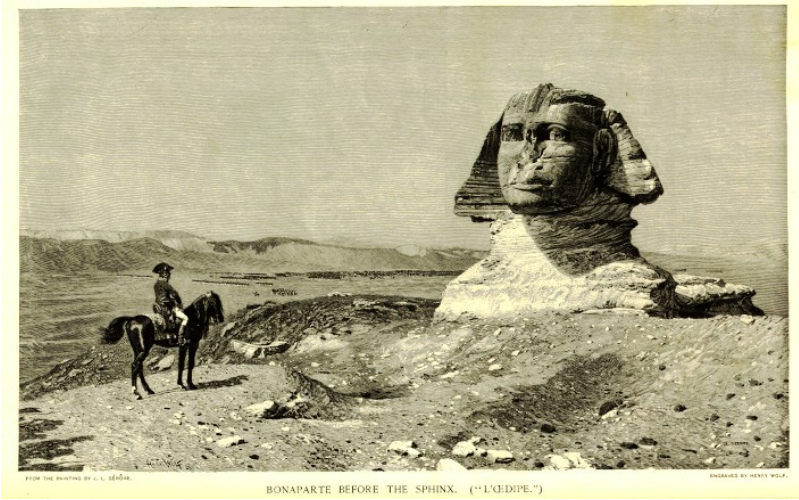
$$|\psi(x_0)\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0+jl\rangle \Longrightarrow \boldsymbol{\rho}=\frac{1}{l}\sum_{x_0=0}^{l-1}|\psi(x_0)\rangle\langle\psi(x_0)|$$

$$\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}\mathbf{T}^k|x_0+jl\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0+k+jl\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0'+jl\rangle$$

$$\mathbf{T}^k\boldsymbol{\rho}\mathbf{T}^{-k}=\boldsymbol{\rho}$$

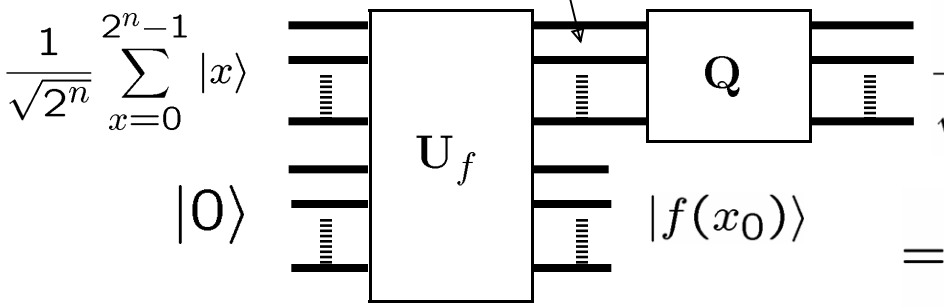
translational invariance $\Longrightarrow$ work in momentum basis

Fourier To The Rescue



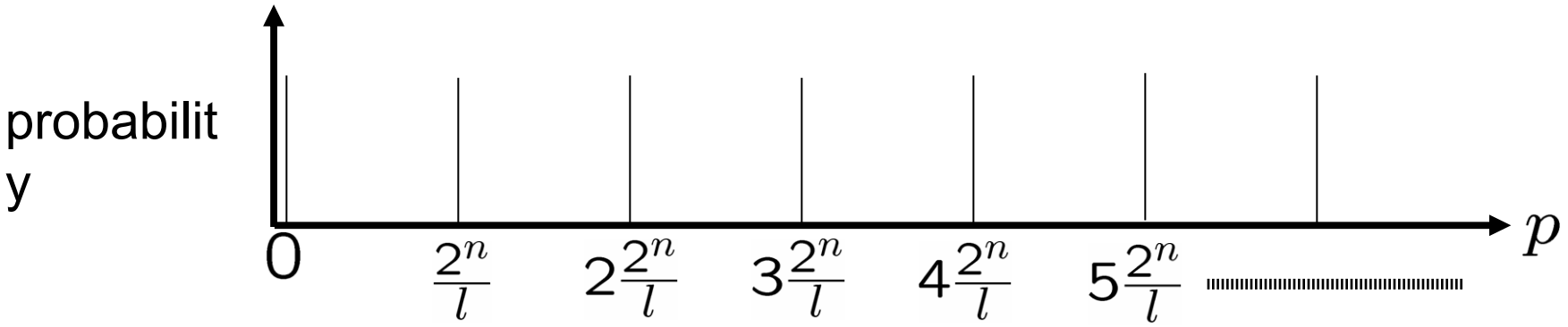
$$\mathbf{Q} = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

$$|\psi(x_0)\rangle = \frac{1}{\sqrt{2^n/l}} \sum_{j=0}^{2^n/l-1} |x_0 + jl\rangle$$



$$\frac{1}{\sqrt{2^{2n}/l}} \sum_{p=0}^{2^n-1} \sum_{j=0}^{2^n/l-1} \exp\left(\frac{2\pi i p(x_0 + jl)}{2^n}\right) |p\rangle$$

$$= \frac{1}{\sqrt{l}} \sum_{p=0|pl=0 \bmod 2^n}^{2^n-1} e^{\frac{2\pi i p x_0}{2^n}} |p\rangle$$



The Wrong Answer

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

“If it were possible to achieve the appropriate superposition of 250 particles, then a quantum computer could perform 10^75 simultaneous computations, completing them in one second! **Oops!**”

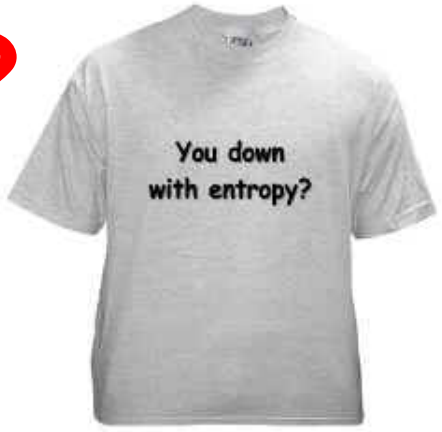
-anonymous web source



n coins

phase space size is 2n

You Down With Entropy?



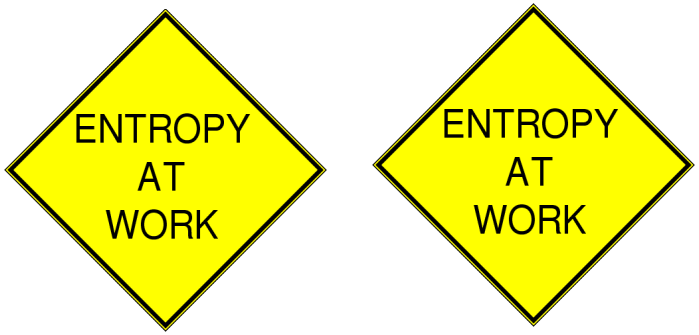
State: $\boldsymbol{\rho} = \frac{1}{l} \sum_{x_0=0}^{l-1} |\psi(x_0)\rangle\langle\psi(x_0)|$

Entropy: $S[\boldsymbol{\rho}] = -\mathrm{Tr}[\boldsymbol{\rho} \log \boldsymbol{\rho}] = \log l$

The quantum algorithm has measured the entropy!

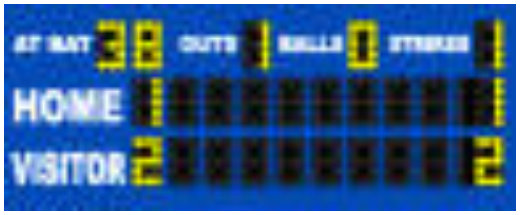


Are quantum computers good at finding the entropy?

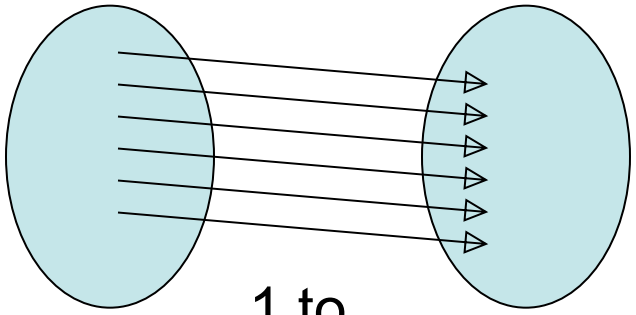


1 to 1 or 2 to 1?





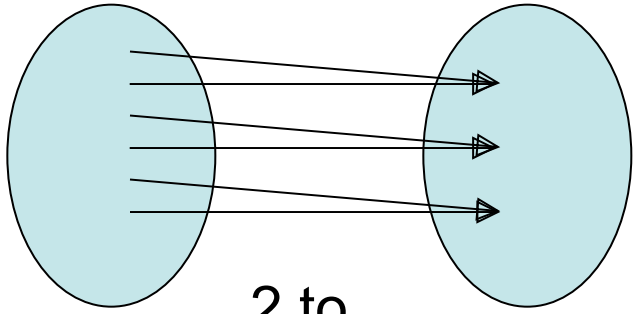
$x \in \{0,1\}^n$ $y \in \{0,1\}^n$



1 to 1

$S[\rho] = n$

$x \in \{0,1\}^n$ $y \in \{0,1\}^n$



2 to 1

$S[\rho] = n - 1$

Theorem (Aaronson 2002): A quantum computer must query f(x) at least 2n/5 (since improved to 2n/3) times to distinguish between these two alternatives.

Experimentalists beware: measuring entropy is HARD!

Useful Symmetry

Today we are engaged in a great search for:

1. Computational problems with symmetry.

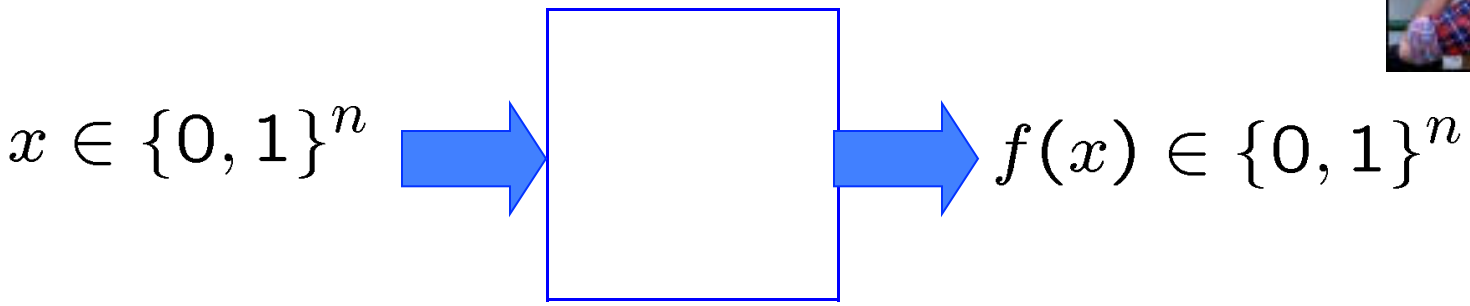


2. Efficient quantum transforms to exploit symmetry.

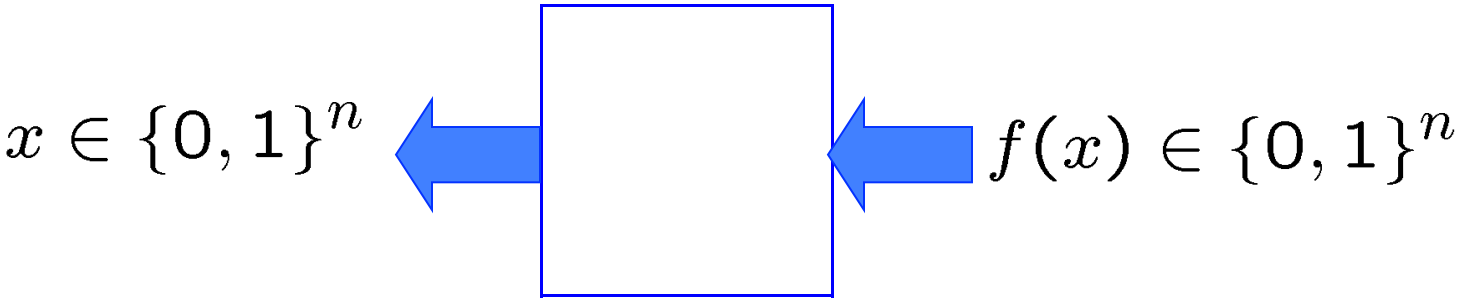


1-way Functions





computing f(x) is easy
(can be done in time polynomial in n)



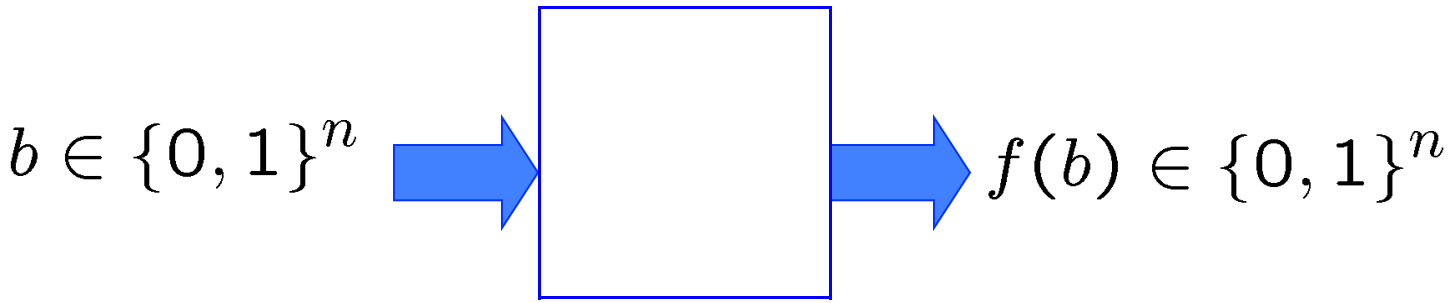
computing f-1(x) is hard
(suspected that it takes a time exponential in n)

1-way Subset Sum

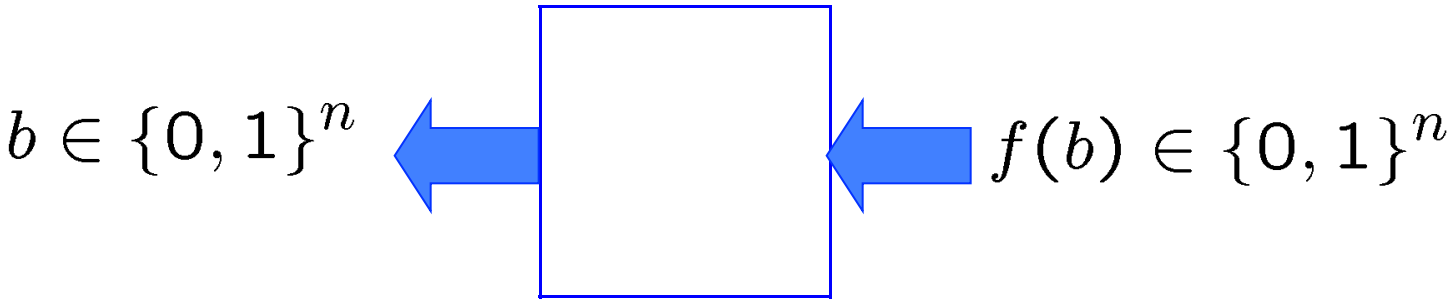
Randomly selected $X_i \in \{0,1\}^n$

$$f(b) = b_1 X_1 + b_2 X_2 + \ldots + b_n X_n \bmod 2^n$$

$b_i \in \{0,1\}$ ith bit of b



easy



har
d

Trapdoor

$x \in \{0,1\}^n$ →  → $f_t(b) \in \{0,1\}^n$

easy

$x \in \{0,1\}^n$ ←  ← $f_t(b) \in \{0,1\}^n$

don't know t: hard
know t: easy

$$f_{p,q}(x) = x^e \bmod N \qquad N = pq \quad p,q \text{ prime}$$

$$e \text{ coprime with } (p-1)(q-1)$$

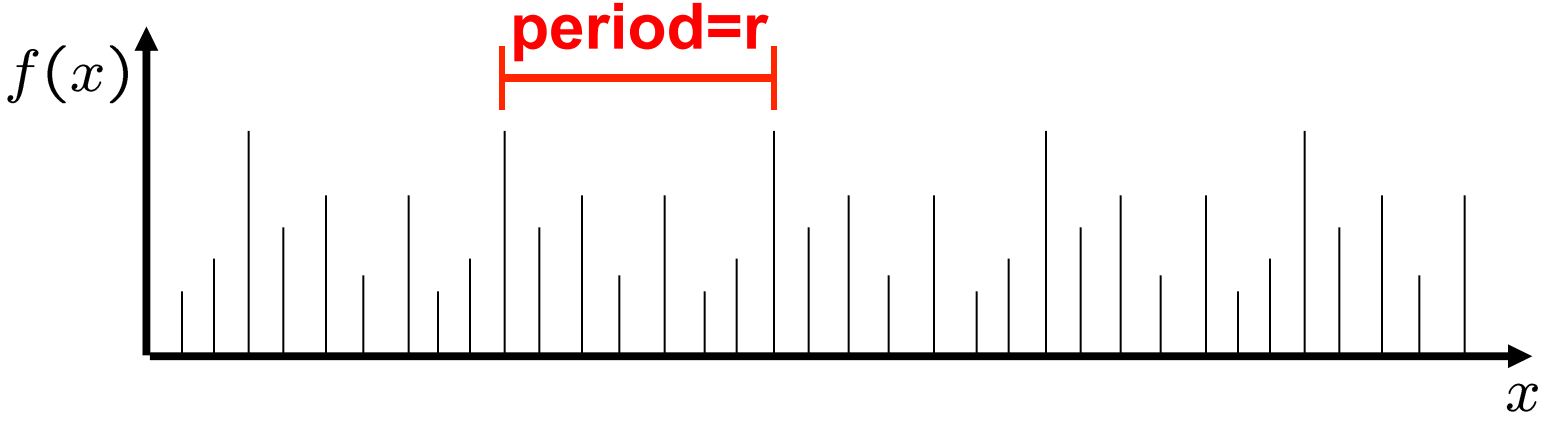
Order Finding

$$f_{p,q}(x) = x^e \bmod N$$

given: x^e, find: x

where's the symmetry?
f not periodic

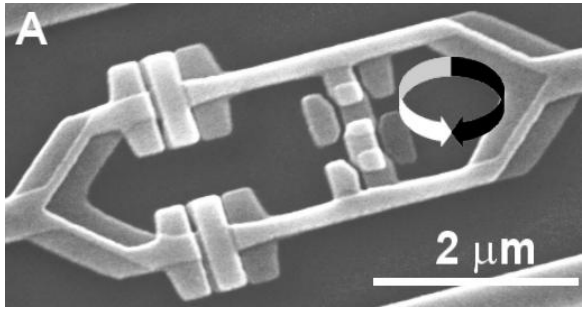
$$x^e, x^{2e}, x^{3e}, x^{4e}, \ldots\ x^{er} = 1$$



$$d = e^{-1} \bmod r \qquad (x^e)^{e^{-1}+kr} = xx^{kr} = x$$

Factoring

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059



efficient order finding
implies
efficient factoring!

3980750864240649373971255005503864911990643623425267084063851895759463889572617685833174727721461074353025362230719730482246329146953020971164598521711305207112563635903975 27

Quantum computers can efficiently factor numbers [Shor 1994]

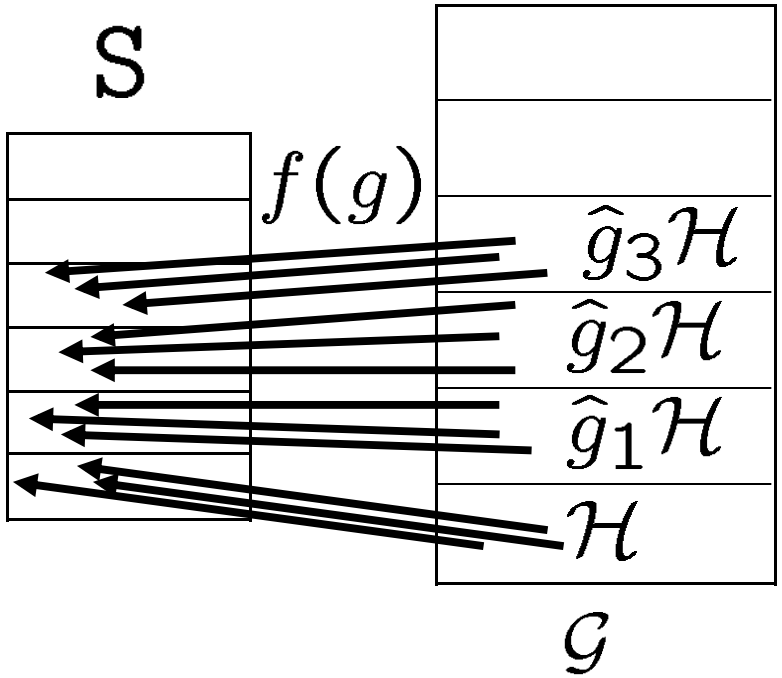
The Hidden Subgroup Problem

Finite group: $\mathcal{G}$ $\quad g_1 g_2 = g_3,\ g_1 g_1^{-1} = e,$ etc.

Given: $f(g)$ $\quad g \in \mathcal{G} \rightarrow \mathrm{S}$

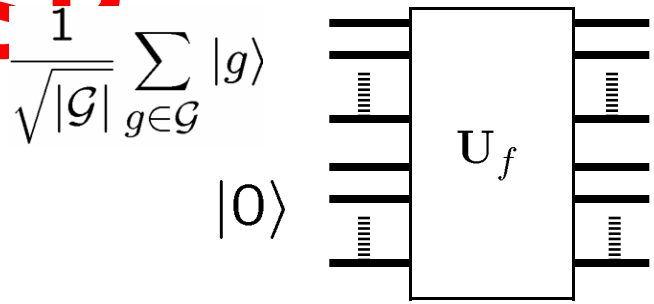
which is constant and distinct on left cosets of some subgroup $\mathcal{H}$

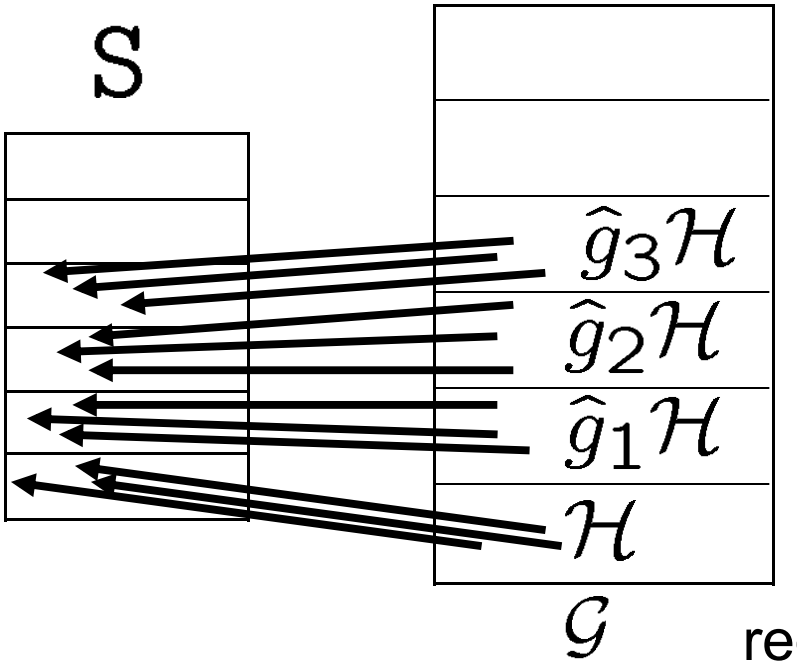
Find: generators for $\mathcal{H}$



$$\hat{g}_3\mathcal{H} = \{\hat{g}_3, \hat{g}_3 h_1, \hat{g}_3 h_2, \ldots\}$$
$$\hat{g}_2\mathcal{H} = \{\hat{g}_2, \hat{g}_2 h_1, \hat{g}_2 h_2, \ldots\}$$
$$\hat{g}_1\mathcal{H} = \{\hat{g}_1, \hat{g}_1 h_1, \hat{g}_1 h_2, \ldots\}$$
$$\mathcal{H} = \{e, h_1, h_2, \ldots\}$$

Symmetry and the HSP





function register $f(\hat{g}_i)$

outcome $|\hat{g}_i\mathcal{H}\rangle = \frac{1}{\sqrt{|\mathcal{H}|}}\sum_{h\in\mathcal{H}}|\hat{g}_i h\rangle$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|}\sum_i |g_i\mathcal{H}\rangle\langle g_i\mathcal{H}|$$

$$\mathbf{D}_L(g)\rho\mathbf{D}_L(g^{-1}) = \rho$$

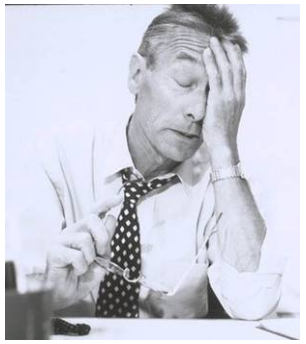
regular representation: $\mathbf{D}_L(g)|g'\rangle = |gg'\rangle$

SYMMETRY!

We Care About HSP Because…

Group $\mathcal{G}$.	classical problem
Abelian	
$\mathbb{Z}_n$	period finding order finding factoring
$\mathbb{Z}_n \times \mathbb{Z}_n$	discrete logarithm
Abelian groups	*efficiently solvable*
	efficiently solvable?
$\mathcal{D}_N$ dihedral group	certain shortest vector in a lattice problems
$\mathcal{S}_N$ symmetric group	graph isomoprhism
non-Abelian	

Frustration !



It is known that if we have $O(log^k|\mathcal{G}|)$ copies of $\rho_{\mathcal{H}}$ then there exists a measurement to find generators of $\mathcal{H}$ [Knill, Ettinger, Hoyer]

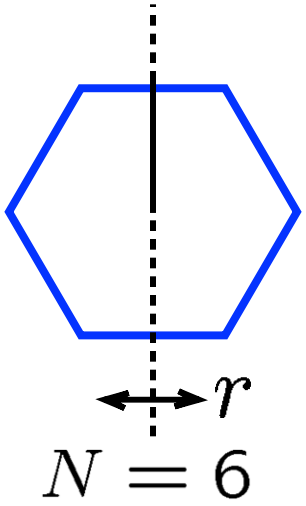
$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}} \otimes \cdots \otimes \rho_{\mathcal{H}}$$

HOWEVER:
the description of this measurement is circuit of depth $O(|\mathcal{G}|^k)$



better description of this measurement?

Dihedral HSP



$\mathcal{D}_N$: symmetry group on N sided regular polygon

$$\mathcal{D}_N = \{r^t s^k, t = 0, 1 \;\; s = 0, 1, \ldots, N-1\}$$

$$r^{t_2} s^{k_2} r^{t_1} s^{k_1} = r^{t_1 + t_2} s^{k_1 + (-1)^{t_1} k_2} \quad r^2 = s^N = e$$

Hidden reflection problem: the hidden subgroup is of the form

$$\mathcal{H}_l = \{e, rs^l\}$$

Best known quantum algorithm [Kuperberg 2003, Regev 2004]

time[Kuperberg] and space[Regev] $2^{O(\sqrt{\log N})}$

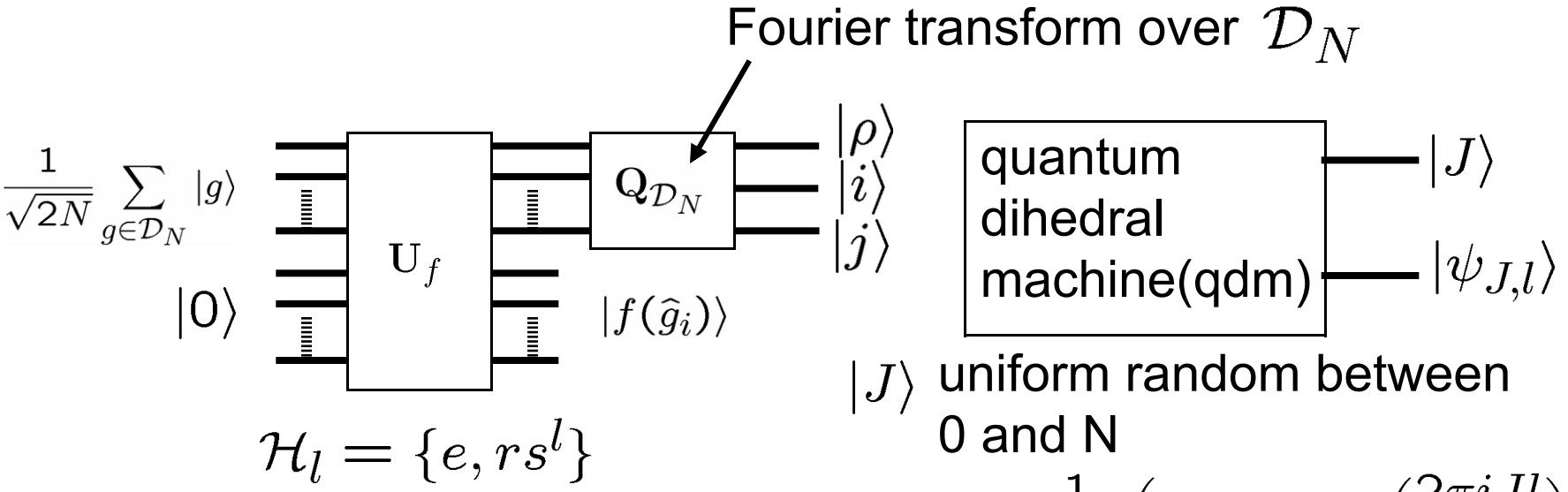
DHSP is important because solving it would solve certain shortest vector in a lattice problems [Regev] (and break a cryptosystem!)

Hidden Phase Problem

symmetry

period finding: $\mathbf{T}\boldsymbol{\rho}\mathbf{T}^{-1} = \boldsymbol{\rho}$ $\mathbf{T}^k$ form an Abelian group

$\mathbf{D}_L(g)\boldsymbol{\rho}\mathbf{D}_L(g^{-1}) = \boldsymbol{\rho}$ $\mathbf{D}_L(g)$ non-Abelian



$\mathcal{H}_l = \{e, rs^l\}$

$|J\rangle$ uniform random between 0 and N

$$|\psi_{J,l}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + \exp\left(\frac{2\pi i J l}{N}\right)|1\rangle\right)$$

Hidden Phase Problem: find l using few queries to the QDM.

1 Use of the QDM

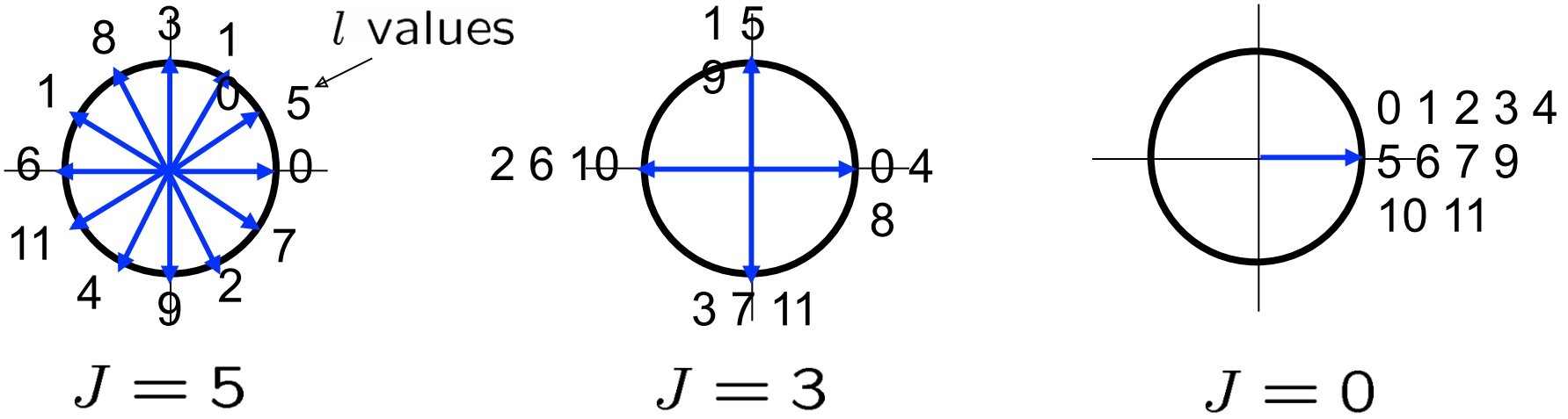
[joint work with Andrew Childs (Caltech) and Wim van Dam (UCSB)]

$$|\psi_{J,l}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + \exp\left(\frac{2\pi i J l}{N}\right)|1\rangle\right)$$

random

$J = 0, 1, \ldots, N-1$

$N = 12$ Examples



$J = 5$ $J = 3$ $J = 0$

What is the optimal measurement for finding the hidden reflection?

(maximize probability of correctly identifying I)

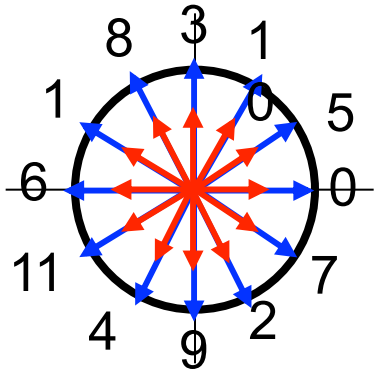
1 Copy Optimal Measurement

$$|\psi_{J,l}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + \exp\left(\frac{2\pi i J l}{N}\right)|1\rangle\right)$$

(assume J≠0)

define non-normalized states

$$|u_{J,k}\rangle = \frac{1}{\sqrt{N}}\left(|0\rangle + \exp\left(\frac{2\pi i J k}{N}\right)|1\rangle\right)$$

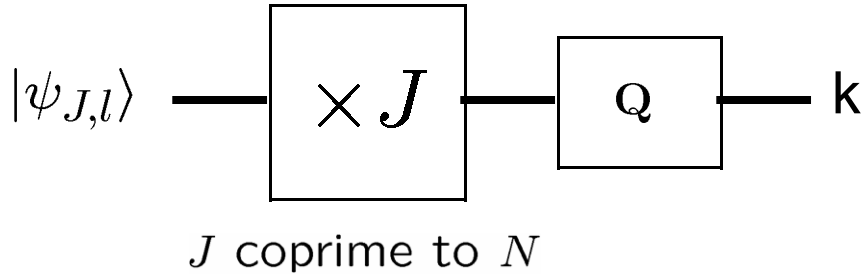


generalized measurement

$$\mathbf{P}_{J,k} = |u_{J,k}\rangle\langle u_{J,k}| \quad \mathbf{P}_{J,N} = \mathbf{I} - \sum_{k=0}^{N-1} \mathbf{P}_{J,k}$$

$$\sum_k \mathbf{P}_{J,k} = \mathbf{I}$$

this measurement is optimal (from Holevo 1973, Ip 2004)



best J (J coprime to N)

$$p_{\text{correct}} = \frac{2}{N}$$

Multicopy Optimal Measurement

q copies

$$|\psi_{J_1,l}\rangle \otimes |\psi_{J_2,l}\rangle \otimes \cdots \otimes |\psi_{J_q,l}\rangle$$

$$= \frac{1}{\sqrt{2^q}} \sum_{\vec{b}} \exp\left(\frac{2\pi i l \vec{b}\cdot\vec{J}}{N}\right) |\vec{b}\rangle \qquad \begin{array}{l} \vec{b} = (b_1, b_2, \ldots, b_q),\ b_i \in \{0,1\} \\ \vec{J} = (J_1, J_2, \ldots, J_q),\ J_i \in \{0, \ldots, N-1\} \end{array}$$

$$|u_{\vec{J},k}\rangle = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} \sum_{\vec{b}|\vec{b}\cdot\vec{J}=t} \frac{1}{\sqrt{\eta_{\vec{J},t}}} \sum \exp\left(\frac{2\pi i k \vec{b}\cdot\vec{J}}{N}\right) |\vec{b}\rangle$$

$\eta_{\vec{J},t}$ is number of $\vec{b}$ such that $\vec{b}\cdot\vec{J} = t$

optimal generalized measurement [Bacon, Childs, van Dam 2004]

$$\mathbf{P}_{\vec{J},k} = |u_{\vec{J},k}\rangle\langle u_{\vec{J},k}|$$

if q = log (N) then implementing this measurement solves DHSP.

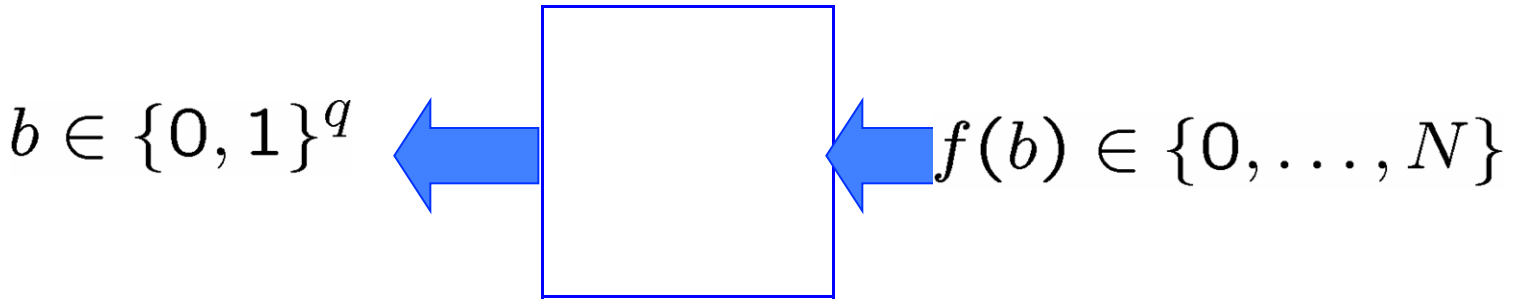
Optimal Measurement and Subset Sum

What makes implementing $\mathbf{P}_{\vec{J},k} = |u_{\vec{J},k}\rangle\langle u_{\vec{J},k}|$ hard?

$$|u_{\vec{J},k}\rangle = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} \sum_{\vec{b}|\vec{b}\cdot\vec{J}=t} \frac{1}{\sqrt{\eta_{\vec{J},t}}} \sum \exp\left(\frac{2\pi i k \vec{b}\cdot\vec{J}}{N}\right) |\vec{b}\rangle$$

subset sum!

$$f(b) = b_1 J_1 + J_2 X_2 + \ldots + b_q J_q \bmod N$$

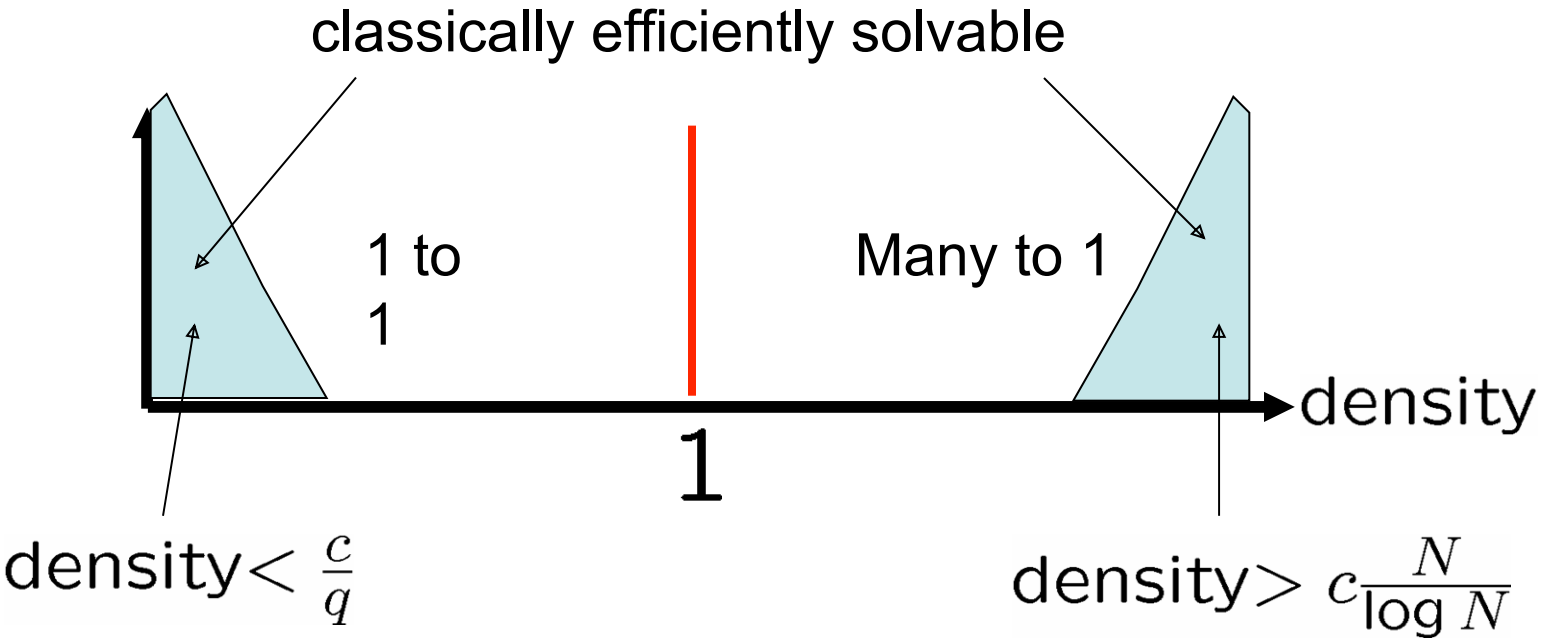


Theorem [Bacon, Childs, van Dam 2004]: If we can implement the measurements $\mathbf{P}_{\vec{J},k} = |u_{\vec{J},k}\rangle\langle u_{\vec{J},k}|$ queries then we can solve subset sum for q numbers of size N.

Subset Sum

$$f(b) = b_1 J_1 + J_2 X_2 + \ldots + b_q J_q \bmod N$$

$$\text{density} = \frac{q}{\log N}$$



efficiently implementing measurement would expand region of solvable subset sum problems!

Review

- Quantum Algorithms Exploit Symmetry
- Symmetric Problems with Applications?
- Optimal Measurements for the Dihedral Hidden Subgroup Problem
- Efficient Implementation of this Measurement Would Solve Certain Subset Sum Problems

A New Hope



Problems a Quantum Computer Might Efficiently Solve

Graph Isomorphism

Restricted Subset Sum Problems

Polynomial Reconstruction

Restricted Shortest Vector in a Lattice Problems

Restricted NP-complete Problems?

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Future CEPI Lectures

Date	Time	Speaker	Topic
11/10/04	reception: 4:15-5 p.m. talk: 5-6 p.m.	Wim van Dam University of California, Santa Barbara	T.B.A.

Internet